Gender Bias and Fairness in Machine Learning-Based Loan Approval

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***Abstract*—This electronic document is a "live" template and already defines the components of your paper [title, text, heads, etc.] in its style sheet. **CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.* (*Abstract*) This paper investigates gender bias and fairness in the machine learning-based loan approval prediction. The synthetic tabular dataset of 45,000 loan applications (20,159 women, 24,841 men) has a class distribution of 77.78% rejected loans and 22.22% approved loans. Four classification models (Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting) are trained and evaluated. A preprocessing workflow implemented using scikit-learn, including feature engineering that expands the dataset to 17 features, is combined with 5-fold stratified cross-validation to assess model performance. Random Forest achieves the best performance among the evaluated models (Accuracy: 0.930, F1: 0.831, ROC-AUC: 0.976). Fairness is evaluated by using Demographic Parity Difference (DPD), Disparate Impact Ratio (DIR), and Equal Opportunity Difference (EOD). The results illustrate a close zero bias on each of the provided models, which is in line with the equal gender approval rates that can be observed in the synthetic dataset (DPD ≤ 0.0072, DIR ≥ 0.969, EOD ≤ 0.009). This study provides a reproducible benchmarking framework for assessing algorithmic fairness in automated loan approval systems.**

I. Introduction

The use of machine learning (ML) models is becoming more widely used in automating financial decisions that have high stakes, such as loan issuance and credit rating. As much as automation has the benefit of increasing processing speed and consistency, the systems can foster or increase the presence of biases that exist within society in the training data, leading to unequal returns for various groups of society members [1]. It is especially catastrophic as gender bias in loan lending may limit access to home ownership, loans to finance education, and capital to start businesses, having socioeconomic implications over the long term [5].

It has been shown that even the unequal treatment between male and female borrowers revealed by historical credit statistics, when given by the ML classifier that is trained on historical data, may result in uneven approvals of the loans [3]. The major issue is that gender bias may be created indirectly, using correlated proxy variables, like income, work stability, or even the purpose of a loan, without assigning gender as an explicit model factor. This is a phenomenon called proxy discrimination, which highlights the importance of post hoc fairness analysis.

This paper investigates whether ML classifiers trained on a synthetic loan approval dataset demonstrate gender-based disparities in prediction outcomes, and whether these disparities can be measured using conventional fairness measures. The prediction problem is binary classification, where the sample size of 13 characteristics of applicants predicts the loan status (approved = 1 , rejected = 0). Gender becomes a shielded characteristic, out of model training, and is utilized merely in evaluating fairness.

The work will yield: (1) a fully operationalized, reusable ML pipeline in Python using scikit-learn; (2) feature engineering that is domain-aware, increasing the feature space to 17 columns; (3) systematic performance evaluation of four classifier families based on their predictive and gender fairness; and (4) a validated fairness evaluation framework (DPD, DIR, EOD) as a baseline for future bias-mitigation phases.

II. Related work

The literature on bias, fairness, and discrimination in automated credit and loan systems has been widely investigated in recent years. This section reviews six recent studies (2024–2025) that are closely related to this work.

Thu et al. [1] compare fairness-aware machine learning models with traditional classifiers, including Logistic Regression, Random Forest, and Support Vector Machines, on benchmark credit datasets. They consider fairness metrics such as Demographic Parity Difference and Equalized Odds, and they show that conventional classifiers create large gender disparities, while fairness-sensitive approaches lessen but do not fully abolish these disparities. The fairness metrics used in their work inspired the metrics used in the present study.

Chen et al. [2] introduce a framework of fairness-based measurement of ML-based credit rating based on Shapley-Lorenz decomposition and tested on approximately 119,857 records of small and medium enterprises in various European countries. They show that the impact of fairness is extremely sensitive to the characteristics of the datasets, such as class imbalance, correlation in features, and sample size, highlighting the significance of thorough exploratory data analysis (EDA).

Liu et al. [3] determine the equality of credit scoring systems between the genders by evaluating the mis-calibration effects between gender groups using the large-scale alternative and traditional credit data. Their results reveal a systematic

overestimation of predicted default risk for female borrowers over male equivalents with the same observed risk characteristics, directly testing the point of gender discrimination in credit systems between algorithms.

Abbas [4] investigates fairness in a credit decision pipeline using information-theoretic analysis on 5000 anonymized US loan applications to learn a Random Forest classifier. Equity is measured by comparing approval rates to subgroup demographics by using AUC as the main performance metric. The research indicates that bias in the approval algorithm can still remain even when gender is not a direct input, and this indicates the problem of proxy discrimination.

Oluoch and Alhassan [5] focus on the gender credit gap phenomenon in low-income countries, and they analyze the large-scale survey data with the help of econometric regression analysis. Although the study is not focused on machine learning, its results show structural disadvantages that women face in accessing credit. The biases revealed suggest that ML systems trained on similar data may inherit this bias.

Ayari et al. [6] provide a synthesis of the systematic literature review that comprises 63 peer-reviewed studies on ML credit scoring published between 2018 and 2024. They record that the family of models that is predominant includes the following: Logistic Regression, Random Forest, Gradient Boosting, Decision Trees, and Neural Networks, with accuracy, AUC, F1-score, precision, and recall being one of the most common evaluation metrics. The modeling selection and evaluation strategy in this study was directly informed by this review.

III. DATASET, PREPROCESSING, AND FEATURE ENGINEERING

A. Dataset Characterization

The data set on which the study is based is loan_data.csv, a simulated tabular dataset containing 45,000 loan applications. It contains 14 attributes, comprising 13 input features and 1 binary target value representing loan status (approved = 1, rejected = 0). The dataset consists of 20,159 female applicants (44.8%) and 24,841 male applicants (55.2%). The target class distribution is imbalanced, with 35,000 rejections (class 0, 77.78%) and 10,000 approvals (class 1, 22.22%). The dataset has no missing values, and all records are complete. A validity check identified a small number of outliers, including 7 records where the person's age surpasses 100. These outliers were retained to maintain dataset integrity.

Key numerical features exhibit wide variability. Annual income ranges from $8,000 to $7,200,766 (mean $80,319), credit scores range from 390 to 850 (mean 632.6), loan amounts range from $500 to $35,000 (mean $9,583), and interest rates range from 5.42% to 20.00% (mean 11.01%). Categorical features include loan intent, education level, home ownership, and prior default history. Loan intent categories include Education (9,153), Medical (8,548), Venture (7,819), Personal (7,552), Debt Consolidation (7,145), and Home Improvement (4,783). Education levels include Bachelor (13,399), Associate (12,028), High School (11,972), Master (6,980), and Doctorate (621). Home ownership categories include Rent (23,443), Mortgage (18,489), Own (2,951), and Other (117). Prior default history is almost balanced, with 22,858 applicants (50.8%) having prior defaults and 22,142 (49.2%) having none. Table I shows key features of some selected features with their types and ranges.

TABLE I. KEY DATASET FEATURES AND STATISTICS

Feature	Type	Range / Values	Role
person_age	Num.	20–144 (mean 27.8)	Input feature
person_income	Num.	$8,000–$7,200,766	Input feature
credit_score	Num.	390–850 (mean 632.6)	Input feature
loan_amnt	Num.	$500–$35,000	Input feature
loan_int_rate	Num.	5.42%–20.00%	Input feature
loan_percent_income	Num.	0.0–1.0	Input feature
person_gender	Cat.	male/female	Protected attribute
loan_status	Bin.	0 (77.78%) / 1 (22.22%)	Target variable

B. Data Cleaning and Preprocessing

The consistency and data leakage were controlled by applying a single preprocessing workflow with ColumnTransformer and Pipeline in scikit-learn. Portions of eight numerical variables (person age, person income, employment experience, loan amount, loan interest rate, loan percentage of income, credit history length, and credit score) were processed using mean imputation to treat missing values and then normalized with StandardScaler (mean = 0, standard deviation = 1). The scaling of all numerical features makes this standardization vital, and is important especially when some features are sensitive to feature magnitude in models.

Six categorical features (person gender, education level, home ownership status, loan intent, loan defaults on file, engineered credit score category) were processed using mode imputation and encoded with OneHotEncoder with the parameter drop=first to decrease multicollinearity while maintaining the essential categorical information. All of the preprocessing stages were integrated into a single workflow so that the same transformations were applied consistently to both training and test data, to prevent data leakage.

The dataset was divided into training and test sets according to the ratio 80/20 with stratify=y and random state=42, resulting in 36,000 training samples and 9,000 test samples. Stratification ensured that in both subsets, the same class structure (77.78%/22.22) was maintained and that the models could be evaluated without bias based on the imbalanced classes.

C. Exploratory Data Analysis

The 77.78%/22.22% imbalance in the classification made by exploratory data analysis (EDA) strongly demonstrated that the dataset was mostly biased in favor of a single category. Due to this imbalance, accuracy would not be a reliable criterion to measure a model, since it can reach high accuracy by just classifying the majority. Hence, alternative measures of evaluation like F1-score and ROC-AUC were used as they would be a more accurate representation of

model performance under imbalanced data sets and give a better balance between precision and recall.

Approval rates were almost the same between the genders, with the females at 22.25% (4,485 out of 20,159) and the males at 22.20% (5,515 out of 24,841). This is a very small difference of 0.05 percent, indicating that there is not much difference between genders on a raw basis.

Correlation analysis showed that loan percent income, credit score, and pre-existing loan defaults were most correlated with the target variable, which means that they are crucial in determining whether a loan is approved or not. There was also outlier detection by the use of the IQR that helped to identify extreme values of the variables, especially age and income. Such outliers were left to maintain realistic variation in the data.

In addition, the histograms of the income distribution were highly right-skewed with a maximum income of $7.2 million as compared to the mean of 80,000. This was what made the use of logarithmic transformation a way of normalizing data so that the model performance can be enhanced.

D. Feature Engineering

Four domain-informed features were engineered, and the number of features in the feature matrix was increased to 17 columns. Each of the engineered features is described in Table II.

Engineered Feature	Type	Definition and Rationale
log_income	Numeric	np.log1p(person_income) — reduces right skew in income; stabilizes variance for linear models
high_debt_ratio	Binary	1 if loan_percent_income > 0.30 — flags excessive debt burden per standard underwriting thresholds
employment_stability	Binary	1 if person_emp_exp > 5 years — captures long-term employment as a proxy for income security
credit_score_category	Ordinal	FICO bands: poor (<580), fair (580–670), good (670–740), excellent (>740) — captures nonlinear credit risk

The thresholds used for credit score categorization follow standard FICO score ranges. The 30% high-debt ratio is in line with the traditional debt-to-income ratios commonly employed in the underwriting of consumer lending and borrowing. Logarithmic transformation is applied to income to address skewness and improve model performance. The binary features (high debt ratio, employment stability) are coded as numeric in the pipeline they are scaled using StandardScaler.

IV. METHODOLOGY

This study examines gender bias and fairness in machine learning-based loan approval using a synthetic tabular dataset. The task is a binary classification problem in which the model predicts whether a loan application is approved or rejected. The dataset contains 45,000 loan applications with both numerical and categorical features, making it suitable for supervised learning and fairness analysis. Because the dataset is synthetic, the findings are interpreted as evidence of model behavior under simulated conditions rather than direct evidence of real-world lending discrimination.

Four baseline models were implemented: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. These models were selected to provide a meaningful comparison between a linear baseline, a single tree model, and ensemble methods. The goal of this phase was to reproduce strong baseline models under a consistent experimental setting rather than to perform extensive hyperparameter tuning.

To improve data representation, four engineered features were added. The first is log_income, which applies a logarithmic transformation to income in order to reduce skewness. The second is high_debt_ratio, a binary feature that marks applicants whose loan-to-income ratio is greater than 30%. The third is employment_stability, which indicates whether an applicant has more than five years of employment experience. The fourth is credit_score_category, which groups credit scores into broader creditworthiness levels. These features were designed using insights from exploratory analysis and domain knowledge from lending and credit scoring.

A unified preprocessing pipeline was used for all models. Numerical and categorical features were identified automatically according to their data types. Numerical features were processed using mean imputation followed by standardization with StandardScaler. Categorical features were processed using mode imputation followed by one-hot encoding with handle_unknown='ignore'. This workflow was implemented using Pipeline and ColumnTransformer to ensure consistency and prevent data leakage.

For evaluation, the dataset was split into 80% training data and 20% testing data using stratification on the target variable. In addition, 5-fold stratified cross-validation was applied to provide a more reliable estimate of model generalization. Since the dataset is imbalanced, the models were evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Fairness was also evaluated across gender groups using Demographic Parity Difference (DPD), Disparate Impact Ratio (DIR), and Equal Opportunity Difference (EOD). In general, lower DPD and EOD values and a DIR closer to 1 indicate fairer treatment across groups.

V. EXPERIMENTAL RESULTS

A. Model Performance

The baseline performance comparison shows that Random Forest achieved the strongest predictive results across all reported metrics, followed by Gradient Boosting. Logistic Regression provided stable performance as a simple baseline, while Decision Tree achieved the weakest overall results.

This pattern is consistent with the expectation that ensemble methods are better suited to structured financial data than single-model baselines. The final test-set performance is summarized in Table III.

TABLE III. KEY DATASET FEATURES AND STATISTICS

Model	Accuracy	F1-score	ROC-AUC
Logistic Regression	0.903	0.778	0.957
Decision Tree	0.898	0.772	0.855
Random Forest	0.930	0.831	0.976
Gradient Boosting	0.923	0.810	0.970

B. Cross-Validation Results

The cross-validation results support the same conclusion. Random Forest again achieved the highest average F1-score, followed by Gradient Boosting. The similarity between test-set performance and cross-validation performance suggests that the best models generalize well and are not relying on a single favorable split.

TABLE IV. CROSS-VALIDATION RESULTS

Model	Average F1-score
Logistic Regression	0.768
Decision Tree	0.773
Random Forest	0.828
Gradient Boosting	0.810

Detailed evaluation based on classification reports and confusion matrices shows that all models perform better on the majority class than on the minority class. This is expected given the class imbalance in the dataset, where roughly 77.78% of samples belong to class 0 and 22.22% belong to class 1. Random Forest and Gradient Boosting provide the best balance between precision and recall, while Decision Tree is more vulnerable to overfitting and performance instability. Gradient Boosting, for example, achieved high precision but lower recall, indicating that it misses a portion of true positives even though its positive predictions are often correct.

C. Fairness Evaluation

In addition to predictive performance, fairness across gender groups was evaluated using established model-level fairness metrics. The fairness results show very low disparity across all four models. Demographic Parity Difference and Equal Opportunity Difference are close to 0, while Disparate Impact Ratio remains close to 1. This means the models give very similar outcomes for male and female applicants, with little evidence of gender bias in this dataset.

Among the evaluated models, Logistic Regression showed the most balanced fairness results. Random Forest, however, gave the best overall balance between predictive performance and fairness, since it achieved the highest prediction quality while still maintaining low disparity across gender groups.

TABLE V. FAIRNESS RESULTS

Model	Demographic Parity Difference (DPD)	Disparate Impact Ratio (DIR)	Equal Opportunity Difference (EOD)
Logistic Regression	0.000150	1.000699	0.002103
Decision Tree	0.007194	0.968515	0.000031
Random Forest	0.002511	0.986983	0.006123
Gradient Boosting	0.005812	0.970142	0.008965

Since the dataset is synthetic, these fairness results reflect model behavior under simulated conditions and may not fully represent real-world bias.

VI. DISCUSSION

The results show that good preprocessing and feature engineering play an important role in this task. The engineered variables improved how the dataset represents financial and applicant information, while the preprocessing pipeline ensured that all models were trained and tested under the same conditions. This made the comparison between models more reliable.

A clear pattern in the results is that ensemble methods outperform simpler models. Random Forest was the strongest model across all performance metrics and remained the best under cross-validation. Gradient Boosting also performed well, although its recall was slightly lower. Logistic Regression was not the strongest model in terms of raw performance, but it remained useful because it was stable, interpretable, and had the most balanced fairness profile. Decision Tree performed worst overall, which is likely due to overfitting and weaker generalization.

The fairness results are an important part of the project because the dataset includes gender and the project specifically studies gender bias in loan approval prediction. Even though the earlier data analysis showed similar outcome distributions for male and female applicants, model-level fairness still needed to be checked. The final results show that strong predictive performance can be achieved without creating large fairness disparities. This is one of the main findings of the study.

At the same time, the scope of the conclusions should be kept clear. Because the dataset is synthetic, the results should not be interpreted as proof about real-world lending systems. Instead, they should be understood as a controlled comparison of how different machine learning models behave in terms of both performance and fairness.

VII. CONCLUSION AND FUTURE WORK

This study evaluated four machine learning models for loan approval prediction while also examining fairness across gender groups. The results show that Random Forest achieved the best predictive performance, while Logistic Regression showed the most balanced fairness profile. Cross-validation confirmed that the best-performing models are stable and generalize well. Overall, the findings show that strong prediction performance can be achieved without sacrificing fairness in this dataset.

There are several directions for future work. First, fairness mitigation methods can be tested to see whether disparity can be reduced even further.

Second, the same framework should be applied to real-world lending datasets to test whether the same conclusions still hold outside a synthetic setting. Third, hyperparameter tuning may improve the balance between recall, overall performance, and fairness. Finally, additional fairness metrics and subgroup analyses could provide a broader view of how these models behave across different applicant groups.

VIII. Code Availability

The full implementation, dataset, and reproducible pipeline are available at: https://github.com/Layanx1/ML-loan-approval-fairness-project

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